National Higher School of Advanced Technologies

Academic year: 2026–2027 | First-year master's degree

Machine learning — Practical session 3

Hypothesis spaces, feature representations and model generalization

Objective

Explore how different machine learning models represent relationships in data, investigate how model complexity and feature representation affect learning performance, and evaluate the ability of models to generalize to unseen observations.

Exercise: Study hours and exam scores

Consider the following dataset containing the number of study hours and the corresponding exam scores of ten students.

Student	Study Hours (x)	Exam Score (y)
S1	1	42
S2	2	48
S3	3	55
S4	4	59
S5	5	66
S6	6	70
S7	7	78
S8	8	81
S9	9	89
S10	10	91

The objective is to predict students' exam scores based on their study hours using different regression models.

Part A : Exploring different model representations

1. Visualize the dataset

- Create a scatter plot representing study hours against exam scores.
- Add appropriate axis labels and a title.
- Observe the relationship between the two variables.

2. Fit different regression models

Use predefined Scikit-learn functions to train the following models:

- Linear Regression.
- Polynomial Regression (degree 2).
- Polynomial Regression (degree 3).

For polynomial regression, use PolynomialFeatures to transform the input features before applying Linear Regression.

3. Compare the models

- Plot the predictions of the three models together with the original observations.
- Observe how the fitted curves differ.
- Compare their training performance using RMSE and R^2.

Record your results in the following table:

Model	**Training RMSE**	**Training R^2**
Linear Regression		
Polynomial Regression (degree 2)		
Polynomial Regression (degree 3)		

Part B : Investigating model complexity and generalization

Increasing model complexity can improve the fit to training observations, but it does not necessarily improve predictions on unseen data.

1. Explore different polynomial degrees

Train polynomial regression models using the following degrees:

- Degree 1
- Degree 2
- Degree 3
- Degree 5
- Degree 9

Use the same dataset for all experiments.

2. Evaluate generalization

Since the dataset contains only ten observations, use Leave-One-Out Cross-Validation (LOOCV) to evaluate the models.

Use predefined Scikit-learn functions:

- LeaveOneOut()
- cross_val_score()

Evaluate the models using negative Mean Squared Error and convert the results into RMSE.

3. Analyze the results

Complete the following table:

Polynomial Degree	Training RMSE	LOOCV RMSE
1		
2		
3		
5		
9		

- Compare training RMSE and LOOCV RMSE.
- Identify the model with the lowest LOOCV RMSE.
- Observe how model complexity affects training performance and generalization.
- Plot the LOOCV RMSE against polynomial degree.

Part C : Investigating feature representation and regularization

A machine learning model learns from the features used to represent the data. Different feature representations can influence its ability to capture relationships and generalize.

1. Compare feature representations

Use the following representations:

- Original representation: study hours (x).
- Polynomial representation: x, x^2 and x^3.
- Higher-degree polynomial representation: x, x^2, ..., x^5.

Compare the predictive performance of the resulting models using LOOCV.

2. Investigate regularization

Apply two regularized regression models to the polynomial features:

- Ridge Regression (L2 regularization).
- Lasso Regression (L1 regularization).

Use the predefined Scikit-learn implementations:

- Ridge()
- Lasso()

Use a Scikit-learn Pipeline to combine feature transformation, feature scaling and regression. This ensures that preprocessing is performed correctly within each cross-validation fold.

Test a few regularization strengths (alpha), for example:

- 0.01
- 0.1
- 1
- 10

Evaluate the models using LOOCV RMSE.

3. Compare the results

Complete the following table:

Model	**Feature Degree**	**Regularization**	**LOOCV RMSE**
Linear Regression	1	None	
Polynomial Regression	3	None	
Polynomial Regression	5	None	
Ridge Regression	5	L2	
Lasso Regression	5	L1	

Identify whether regularization improves the predictive performance of polynomial regression.

Part D : Final model comparison

Compare the best-performing models from the previous experiments.

Your comparison should include:

- Linear Regression.
- Polynomial Regression.
- Ridge Regression.
- Lasso Regression.

For each model:

1. Report the training RMSE.

2. Report the LOOCV RMSE.
3. Compare the fitted predictions with the original observations.
4. Identify the model that provides the best balance between fitting the available data and generalizing to unseen observations.

Present your final results in a summary table and a comparative plot.

Deliverables

Submit a Python code containing:

1. Dataset exploration.
2. Implementation using predefined Scikit-learn functions.

Function / Class	Purpose
`LinearRegression()`	Create a linear regression model.
`PolynomialFeatures()`	Generate polynomial features to model nonlinear relationships.
`Ridge()`	Create a Ridge regression model using L2 regularization.
`Lasso()`	Create a Lasso regression model using L1 regularization.
`DecisionTreeRegressor()`	Create a decision tree regression model.
`RandomForestRegressor()`	Create a random forest regression model using multiple decision trees.
`SVR()`	Create a support vector regression model.
`KNeighborsRegressor()`	Create a k-nearest neighbors regression model.

3. Training and cross-validation results.
4. Comparative plots.
5. Final summary table.
6. A short conclusion discussing the influence of model complexity, feature representation and regularization on generalization performance.

Important: All models must be evaluated using the same dataset and the same cross-validation strategy. For preprocessing operations such as feature scaling, use a Pipeline to avoid data leakage.